

3.3.4 Householder QR factorization algorithm



03.3.4 Householder QR factorization, part 1

fit width



Let A be an $m \times n$ with $m \geq n$. We will now show how to compute $A \rightarrow QR$, the QR factorization, as a sequence of Householder transformations applied to A , which eventually zeroes out all elements of that matrix below the diagonal. The process is illustrated in [Figure 3.3.4.1](#).

Original matrix	$\left[\begin{pmatrix} \rho_{11} \\ u_{21} \end{pmatrix}, \tau_1 \right] = \text{Housev} \begin{pmatrix} \alpha_{11} \\ a_{21} \end{pmatrix}$	$\begin{pmatrix} \alpha_{11} & a_{12}^T \\ a_{21} & A_{22} \end{pmatrix} := \begin{pmatrix} \rho_{11} & a_{12}^T - w_{12}^T \\ 0 & A_{22} - u_{21} w_{12}^T \end{pmatrix}$	“Move forward”
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Figure 3.3.4.1. Illustration of Householder QR factorization.

In the first iteration, we partition

$$A \rightarrow \begin{pmatrix} \alpha_{11} & a_{12}^T \\ a_{21} & A_{22} \end{pmatrix}.$$

Let

$$\left[\begin{pmatrix} \rho_{11} \\ u_{21} \end{pmatrix}, \tau_1 \right] = \text{Housev} \left(\begin{pmatrix} \alpha_{11} \\ a_{21} \end{pmatrix} \right)$$

be the Householder transform computed from the first column of A . Then applying this Householder transform to A yields

$$\begin{aligned} \begin{pmatrix} \alpha_{11} & a_{12}^T \\ a_{21} & A_{22} \end{pmatrix} &:= \left(I - \frac{1}{\tau_1} \begin{pmatrix} 1 \\ u_{21} \end{pmatrix} \begin{pmatrix} 1 & u_{21}^T \end{pmatrix}^H \right) \begin{pmatrix} \alpha_{11} & a_{12}^T \\ a_{21} & A_{22} \end{pmatrix} \\ &= \begin{pmatrix} \rho_{11} & a_{12}^T - w_{12}^T \\ 0 & A_{22} - u_{21} w_{12}^T \end{pmatrix}, \end{aligned}$$

where $w_{12}^T = (a_{12}^T + u_{21}^H A_{22})/\tau_1$. Computation of a full QR factorization of A will now proceed with the updated matrix A_{22} .

03.3.4 HQR factorization algorithm, part 2

fit width



Homework 3.3.4.1. Show that

$$\left(\begin{array}{c|c} I & 0 \\ \hline 0 & I - \frac{1}{\tau_1} \begin{pmatrix} 1 \\ u_{21} \end{pmatrix} \begin{pmatrix} 1 \\ u_{21} \end{pmatrix}^H \end{array} \right) = \left(I - \frac{1}{\tau_1} \begin{pmatrix} 0 \\ 1 \\ u_{21} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ u_{21} \end{pmatrix}^H \right).$$

▼ [Solution](#)

$$\begin{aligned} \left(\begin{array}{c|c} I & 0 \\ \hline 0 & I - \frac{1}{\tau_1} \begin{pmatrix} 1 \\ u_{21} \end{pmatrix} \begin{pmatrix} 1 \\ u_{21} \end{pmatrix}^H \end{array} \right) &= I - \left(\begin{array}{c|c} 0 & 0 \\ \hline 0 & \frac{1}{\tau_1} \begin{pmatrix} 1 \\ u_{21} \end{pmatrix} \begin{pmatrix} 1 \\ u_{21} \end{pmatrix}^H \end{array} \right) \\ &= I - \frac{1}{\tau_1} \left(\begin{array}{c|c} 0 & 0 \\ \hline 0 & \begin{pmatrix} 1 \\ u_{21} \end{pmatrix} \begin{pmatrix} 1 \\ u_{21} \end{pmatrix}^H \end{array} \right) \\ &= I - \frac{1}{\tau_1} \left(\begin{array}{c|c} 0 & 0 & 0 \\ \hline 0 & 1 & u_{21}^H \\ 0 & u_{21} & u_{21} u_{21}^H \end{array} \right) \\ &= \left(I - \frac{1}{\tau_1} \begin{pmatrix} 0 \\ 1 \\ u_{21} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ u_{21} \end{pmatrix}^H \right). \end{aligned}$$

More generally, let us assume that after k iterations of the algorithm matrix A contains

$$A \rightarrow \left(\begin{array}{c|c} R_{TL} & R_{TR} \\ \hline 0 & A_{BR} \end{array} \right) = \left(\begin{array}{c|c} R_{00} & r_{01} & R_{02} \\ \hline 0 & \alpha_{11} & a_{12}^T \\ 0 & a_{21} & A_{22} \end{array} \right),$$

where R_{TL} and R_{00} are $k \times k$ upper triangular matrices. Let

$$\left[\begin{pmatrix} \rho_{11} \\ u_{21} \end{pmatrix}, \tau_1 \right] = \text{Housev} \left(\begin{pmatrix} \alpha_{11} \\ a_{21} \end{pmatrix} \right).$$

and update

$$\begin{aligned} A &:= \left(\begin{array}{c|c} I & 0 \\ \hline 0 & I - \frac{1}{\tau_1} \begin{pmatrix} 1 \\ u_{21} \end{pmatrix} \begin{pmatrix} 1 \\ u_{21} \end{pmatrix}^H \end{array} \right) \left(\begin{array}{c|c} R_{00} & r_{01} & R_{02} \\ \hline 0 & \alpha_{11} & a_{12}^T \\ 0 & a_{21} & A_{22} \end{array} \right) \\ &= \left(I - \frac{1}{\tau_1} \begin{pmatrix} 0 \\ 1 \\ u_{21} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ u_{21} \end{pmatrix}^H \right) \left(\begin{array}{c|c} R_{00} & r_{01} & R_{02} \\ \hline 0 & \alpha_{11} & a_{12}^T \\ 0 & a_{21} & A_{22} \end{array} \right) \\ &= \left(\begin{array}{c|c} R_{00} & r_{01} & R_{02} \\ \hline 0 & \rho_{11} & a_{12}^T - w_{12}^T \\ 0 & 0 & A_{22} - u_{21} w_{12}^T \end{array} \right), \end{aligned}$$

where, again, $w_{12}^T = (a_{12}^T + u_{21}^H A_{22}) / \tau_1$.

Let

$$H_k = \left(I - \frac{1}{\tau_1} \begin{pmatrix} 0_k \\ 1 \\ u_{21} \end{pmatrix} \begin{pmatrix} 0_k \\ 1 \\ u_{21} \end{pmatrix}^H \right)$$

be the Householder transform so computed during the $(k+1)$ st iteration. Then upon completion matrix A contains

$$R = \left(\begin{array}{c|c} R_{TL} & \\ \hline 0 & \end{array} \right) = H_{n-1} \cdots H_1 H_0 \hat{A}$$

where $\hat{A}$ denotes the original contents of A and R_{TL} is an upper triangular matrix. Rearranging this we find that

$$\hat{A} = H_0 H_1 \cdots H_{n-1} R$$

which shows that if $Q = H_0 H_1 \cdots H_{n-1}$ then $\hat{A} = QR$.

Typically, the algorithm overwrites the original matrix A with the upper triangular matrix, and at each step u_{21} is stored over the elements that become zero, thus overwriting a_{21} . (It is for this reason that the first element of u was normalized to equal "1".) In this case Q is usually not explicitly formed as it can

be stored as the separate Householder vectors below the diagonal of the overwritten matrix. The algorithm that overwrites A in this manner is given in [Figure 3.3.4.2](#).

